

Geometry & Coordinate Geometry

ANSWER KEY



Distance and midpoint

- 1 Find the distance between (1, 2) and (4, 6).
 $d = \sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9+16} = \sqrt{25} = 5.$
 - 2 Find the midpoint of (2, -3) and (8, 5).
 $\text{Midpoint} = \left(\frac{2+8}{2}, \frac{-3+5}{2} \right) = (5, 1).$
 - 3 Find the distance between (-2, 1) and (3, -11).
 $d = \sqrt{(3-(-2))^2 + (-11-1)^2} = \sqrt{25+144} = \sqrt{169} = 13.$
 - 4 Find the midpoint of (-4, 7) and (6, -3).
 $\text{Midpoint} = \left(\frac{-4+6}{2}, \frac{7-3}{2} \right) = (1, 2).$
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Gradient (slope)

- 5 Find the gradient of the line through (1, 2) and (5, 10).
 $m = \frac{10-2}{5-1} = \frac{8}{4} = 2.$
 - 6 Find the gradient of the line through (-3, 4) and (2, -6).
 $m = \frac{-6-4}{2-(-3)} = \frac{-10}{5} = -2.$
 - 7 Find the gradient of the line through (0, 0) and (6, 9).
 $m = \frac{9-0}{6-0} = \frac{9}{6} = \frac{3}{2}.$
 - 8 State whether the line through (1, 1) and (4, 1) is horizontal, vertical, or neither.
Both points have $y = 1$, so the line is horizontal (gradient = 0).
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Equations of lines

- 9 Find the equation of the line with gradient 2 passing through (3, 5), in the form $y = mx + c$.
 $y - 5 = 2(x - 3) \Rightarrow y = 2x - 1.$
- 10 Find the equation of the line through (0, 4) and (2, 10).
 $m = \frac{10-4}{2-0} = 3.$ Using the y-intercept 4: $y = 3x + 4.$
- 11 Find the x- and y-intercepts of the line $2x + 3y = 12$.
x-intercept ($y = 0$): $x = 6.$ y-intercept ($x = 0$): $y = 4.$

- 12 Write the equation of the line parallel to $y = 4x - 1$ passing through $(2, 9)$.
Same gradient 4: $y - 9 = 4(x - 2) \Rightarrow y = 4x + 1$.
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Parallel and perpendicular lines

- 13 State the gradient of a line perpendicular to a line with gradient 5.
 $m = -\frac{1}{5}$.
- 14 Determine whether the lines $y = 2x + 3$ and $y = 2x - 7$ are parallel, perpendicular, or neither.
Parallel — both have gradient 2.
- 15 Find the equation of the line perpendicular to $y = -\frac{1}{3}x + 2$, passing through $(3, 4)$.
The perpendicular gradient is 3. $y - 4 = 3(x - 3) \Rightarrow y = 3x - 5$.
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The equation of a circle

- 16 Write the equation of a circle centered at the origin with radius 6.
 $x^2 + y^2 = 36$.
- 17 Write the equation of a circle centered at $(2, -3)$ with radius 5.
 $(x - 2)^2 + (y + 3)^2 = 25$.
- 18 Find the radius and center of the circle $(x - 1)^2 + (y + 4)^2 = 49$.
Center: $(1, -4)$. Radius: 7.
- 19 Find the equation of the circle centered at the origin that passes through $(3, 4)$.
Radius $= \sqrt{3^2 + 4^2} = 5$. Equation: $x^2 + y^2 = 25$.
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Circles: chords and arcs

- 20 Points A and B lie on a circle of radius 5 centered at O , with central angle $\angle AOB = 90^\circ$. Find the length of chord AB , using $AB = 2r \sin\left(\frac{\theta}{2}\right)$.
 $AB = 2(5)\sin(45^\circ) = 10(0.7071) \approx 7.07$.
- 21 A chord of a circle with radius 10 cm is 12 cm long. Find the distance from the center to the chord.
Half the chord is 6 cm. Distance $= \sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = 8$ cm.
- 22 Find the circumference of a circle with radius 9 cm, in terms of π .
 $C = 2\pi r = 18\pi$ cm.
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Polygon angle sums

- 23 Find the sum of the interior angles of a hexagon (6 sides).
 $(n - 2)180^\circ = (6 - 2)180^\circ = 720^\circ$.
- 24 Find the measure of each interior angle of a regular pentagon.
Sum $= (5 - 2)180^\circ = 540^\circ$. Each angle $= \frac{540^\circ}{5} = 108^\circ$.
- 25 A polygon has an interior angle sum of 1440° . Find the number of sides.
 $(n - 2)180^\circ = 1440^\circ \Rightarrow n - 2 = 8 \Rightarrow n = 10$.
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Perimeter and area

- 26 A rectangle has length 12 and width 7. Find its perimeter and area.
Perimeter $= 2(12 + 7) = 38$. Area $= 12 \times 7 = 84$.
- 27 A right triangle has legs 6 and 8. Find its area and hypotenuse.
Area $= \frac{1}{2}(6)(8) = 24$. Hypotenuse $= \sqrt{36 + 64} = \sqrt{100} = 10$.
- 28 A regular hexagon has side length 4, with area $A = \frac{3\sqrt{3}}{2}s^2$. Find its area in exact form.
 $A = \frac{3\sqrt{3}}{2}(16) = 24\sqrt{3}$.
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Coordinate geometry: identifying shapes

- 29 Show that the triangle with vertices $A(0, 0)$, $B(4, 0)$, $C(4, 3)$ is a right triangle.
 AB is horizontal and BC is vertical, so they are perpendicular; the right angle is at B .
- 30 Determine whether the points $A(-2, 1)$, $B(2, 4)$, $C(5, 0)$ form a right triangle, using the squared side lengths.
 $AB^2 = 4^2 + 3^2 = 25$. $BC^2 = 3^2 + 4^2 = 25$. $AC^2 = 7^2 + 1^2 = 50$. Since $AB^2 + BC^2 = 50 = AC^2$, the triangle has a right angle at B .
- 31 Determine whether the triangle with vertices $A(1, 2)$, $B(5, 2)$, $C(3, 6)$ is isosceles.
 $AC = \sqrt{2^2 + 4^2} = \sqrt{20}$ and $BC = \sqrt{2^2 + 4^2} = \sqrt{20}$, so $AC = BC$ and the triangle is isosceles.
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The perpendicular bisector of a segment

- 32 Find the equation of the perpendicular bisector of the segment joining $(2, 3)$ and $(8, 7)$.
Midpoint $= (5, 5)$. Segment gradient $= \frac{7-3}{8-2} = \frac{4}{6} = \frac{2}{3}$, so the perpendicular gradient is $-\frac{3}{2}$. Equation:
 $y - 5 = -\frac{3}{2}(x - 5) \Rightarrow y = -\frac{3}{2}x + \frac{25}{2}$.

- 33** Find the equation of the perpendicular bisector of the segment joining $(0, 0)$ and $(6, 8)$.
 Midpoint $= (3, 4)$. Segment gradient $= \frac{8}{6} = \frac{4}{3}$, so the perpendicular gradient is $-\frac{3}{4}$. Equation:
 $y - 4 = -\frac{3}{4}(x - 3) \Rightarrow y = -\frac{3}{4}x + \frac{25}{4}$.
- 34** A line segment has endpoints $(1, 5)$ and $(7, 5)$. Find the equation of its perpendicular bisector.
 Midpoint $= (4, 5)$. The segment is horizontal (gradient 0), so the perpendicular bisector is the vertical line $x = 4$.

Word problems: geometry on a coordinate grid

- 35** This question has two parts. A city park is designed on a coordinate grid (in meters), with two gates at $A(0, 0)$ and $B(80, 60)$.
- a** Find the straight-line distance between the two gates.
 $AB = \sqrt{80^2 + 60^2} = \sqrt{6400 + 3600} = \sqrt{10000} = 100$ meters.
- b** A fountain is placed at the midpoint between the two gates. Find its coordinates. Midpoint
 $= \left(\frac{0+80}{2}, \frac{0+60}{2} \right) = (40, 30)$.
- 36** This question has two parts. A circular irrigation sprinkler is centered at $(5, 5)$ on a farm's coordinate grid (in meters) and has a radius of 12 meters.
- a** Write the equation of the circle covered by the sprinkler. $(x - 5)^2 + (y - 5)^2 = 144$.
- b** Determine whether a tree at $(14, 9)$ lies inside, on, or outside the watered region. Distance from center $= \sqrt{(14 - 5)^2 + (9 - 5)^2} = \sqrt{81 + 16} = \sqrt{97} \approx 9.85$ meters, which is less than the radius 12, so the tree lies inside the watered region.

MCQ — Distance and midpoint

- 37** Find the distance between $(2, 3)$ and $(6, 6)$.

a. 5 ✓

b. 25

c. $\sqrt{7}$

d. 7

$$\sqrt{(6 - 2)^2 + (6 - 3)^2} = \sqrt{16 + 9} = 5.$$

38 Find the midpoint of $(3, -4)$ and $(9, 6)$.

a. $(6, 1)$ ✓

b. $(6, 2)$

c. $(12, 2)$

d. $(3, 10)$

$$\left(\frac{3+9}{2}, \frac{-4+6}{2} \right) = (6, 1).$$

39 Find the distance between $(-3, 2)$ and $(5, -13)$.

a. 17 ✓

b. 15

c. 19

d. 289

$$\sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17.$$

40 Find the midpoint of $(-5, 8)$ and $(7, -2)$.

a. $(6, 1)$

b. $(1, 6)$

c. $(2, 6)$

d. $(1, 3)$ ✓

$$\left(\frac{-5+7}{2}, \frac{8-2}{2} \right) = (1, 3).$$

41 Find the distance between the two points shown.

a. $\sqrt{5}$

b. 4

c. 3

d. 5 ✓

$$\sqrt{(5-1)^2 + (5-2)^2} = \sqrt{16+9} = 5.$$

42 Find the midpoint of the segment joining the two points shown.

a. (4, 3)

b. (2, 6)

c. (2, 3) ✓

d. (8, 6)

$$\left(\frac{-2 + 6}{2}, \frac{3 + 3}{2} \right) = (2, 3).$$

MCQ — Gradient (slope)

43 Find the gradient of the line through (2, 3) and (6, 11).

a. 8

b. 0.5

c. 4

d. 2 ✓

$$m = \frac{11 - 3}{6 - 2} = 2.$$

44 Find the gradient of the line through (−2, 5) and (3, −10).

a. $-\frac{1}{3}$

b. −15

c. −3 ✓

d. 3

$$m = \frac{-10 - 5}{3 - (-2)} = \frac{-15}{5} = -3.$$

45 Find the gradient of the line through (0, 0) and (8, 12).

a. 1.5 ✓

b. $\frac{8}{12}$

c. 96

d. $\frac{2}{3}$

$$m = \frac{12}{8} = 1.5.$$

46 State whether the line through (2, 3) and (2, 9) is horizontal, vertical, or neither.

a. Cannot be determined

b. Neither

c. Horizontal

d. Vertical ✓

Both points share $x = 2$: the line is vertical.

MCQ — Equations of lines

47 Find the equation of the line with gradient 3 passing through (2, 4), in the form $y = mx + c$.

a. $y = 3x + 4$

b. $y = 3x - 2$ ✓

c. $y = 3x + 2$

d. $y = 2x - 3$

$$4 = 3(2) + c \Rightarrow c = -2.$$

48 Find the equation of the line through (0, 5) and (3, 14).

a. $y = 5x + 3$

b. $y = 9x + 5$

c. $y = 3x + 5$ ✓

d. $y = 3x - 5$

$$m = \frac{14 - 5}{3} = 3; c = 5 \text{ (the } y\text{-intercept).}$$

49 Find the x - and y -intercepts of the line $3x + 4y = 24$.

a. $x = 8, y = 6$ ✓

b. $x = 6, y = 8$

c. $x = 24, y = 24$

d. $x = 4, y = 3$

Set $y = 0$: $x = 8$. Set $x = 0$: $y = 6$.

50 Write the equation of the line parallel to $y = 5x - 2$ passing through (1, 8).

a. $y = 5x + 3$ ✓

b. $y = -\frac{1}{5}x + 3$

c. $y = 5x - 3$

d. $y = 5x + 8$

$$8 = 5(1) + c \Rightarrow c = 3.$$

51 Find the equation of the line through (1, 3) and (4, 12).

a. $y = 3x + 3$

b. $y = x + 3$

c. $y = 3x - 3$

d. $y = 3x$ ✓

$$m = \frac{12 - 3}{3} = 3; 3 = 3(1) + c \Rightarrow c = 0.$$

52 The graph shows a line. Find its equation in the form $y = mx + c$ by reading the graph.

a. $y = \frac{1}{2}x + 1$

b. $y = 2x + 1$ ✓

c. $y = x + 1$

d. $y = 2x - 1$

The line crosses the y -axis at 1 and rises 2 for every 1 across: $y = 2x + 1$.

MCQ — Parallel and perpendicular lines

53 State the gradient of a line perpendicular to a line with gradient 4.

a. $-\frac{1}{4}$ ✓

b. 4

c. $\frac{1}{4}$

d. -4

Perpendicular gradients multiply to -1 : $-\frac{1}{4}$.

54 Determine whether the lines $y = 3x + 2$ and $y = 3x - 5$ are parallel, perpendicular, or neither.

a. The same line

b. Neither

c. Perpendicular

d. **Parallel** ✓

Both have gradient 3: parallel.

55 Find the equation of the line perpendicular to $y = -\frac{1}{4}x + 3$, passing through (4, 5).

a. $y = 4x - 5$

b. $y = 4x + 11$

c. $y = -\frac{1}{4}x + 6$

d. $y = 4x - 11$ ✓

Perpendicular gradient = 4: $5 = 4(4) + c \Rightarrow c = -11$.

56 Determine whether the lines $y = 2x + 1$ and $y = -0.5x + 4$ are parallel, perpendicular, or neither.

a. The same line

b. Parallel

c. Neither

d. Perpendicular ✓

Gradients 2 and -0.5 multiply to -1 : perpendicular.

MCQ — The equation of a circle

57 Write the equation of a circle centered at the origin with radius 8.

a. $x^2 + y^2 = 64$ ✓

b. $x^2 + y^2 = 16$

c. $x^2 + y^2 = 8$

d. $(x - 8)^2 + (y - 8)^2 = 64$

$r^2 = 64$.

58 Write the equation of a circle centered at $(3, -2)$ with radius 7.

a. $(x - 3)^2 - (y + 2)^2 = 49$

b. $(x - 3)^2 + (y + 2)^2 = 49$ ✓

c. $(x - 3)^2 + (y + 2)^2 = 7$

d. $(x + 3)^2 + (y - 2)^2 = 49$

$(x - 3)^2 + (y - (-2))^2 = 7^2.$

59 Find the radius and center of the circle $(x + 2)^2 + (y - 5)^2 = 81$.

a. Center $(2, -5)$, radius 9

b. Center $(-2, 5)$, radius 9 ✓

c. Center $(-2, 5)$, radius 81

d. Center $(2, 5)$, radius 9

$(x - (-2))^2 + (y - 5)^2 = 9^2.$

60 Find the equation of the circle centered at the origin that passes through $(6, 8)$.

a. $x^2 + y^2 = 100$ ✓

b. $x^2 + y^2 = 10$

c. $x^2 + y^2 = 196$

d. $x^2 + y^2 = 48$

$r = \sqrt{6^2 + 8^2} = 10; r^2 = 100.$

61 Write the equation of the circle shown, centered at the origin.

a. $x^2 + y^2 = 10$

b. $x^2 + y^2 = 5$

c. $x^2 + y^2 = 20$

d. $x^2 + y^2 = 25$ ✓

$r = 5; r^2 = 25.$

62 The circle shown is centered at $(1, 2)$ with the radius marked. Write its equation.

a. $(x - 1)^2 + (y - 2)^2 = 8$

b. $(x + 1)^2 + (y + 2)^2 = 16$

c. $(x - 1)^2 + (y - 2)^2 = 16$ ✓

d. $(x - 1)^2 + (y - 2)^2 = 4$

$(x - 1)^2 + (y - 2)^2 = 4^2$.

MCQ — Circles: chords and arcs

63 Points A and B lie on a circle of radius 8 centered at O , with central angle $\angle AOB = 60^\circ$. Find the arc length AB , in terms of π .

a. $\frac{16\pi}{3}$

b. $\frac{4\pi}{3}$

c. 16π

d. $\frac{8\pi}{3}$ ✓

Arc $= \frac{60}{360} \times 2\pi(8) = \frac{8\pi}{3}$.

64 A chord of a circle with radius 13 cm is 24 cm long. Find the distance from the center to the chord.

a. 7

b. 1

c. 12

d. 5 ✓

Half-chord $= 12$; distance $= \sqrt{13^2 - 12^2} = \sqrt{25} = 5$ cm.

65 Find the circumference of a circle with radius 11 cm, in terms of π .

a. 121π

b. 44π

c. 22π ✓

d. 11π

$$C = 2\pi(11) = 22\pi \text{ cm.}$$

66 Find the arc length of the 90° sector shown, in terms of π .

a. 12π

b. 3π ✓

c. 1.5π

d. 6π

$$\text{Arc} = \frac{90}{360} \times 2\pi(6) = 3\pi.$$

MCQ — Polygon angle sums

67 Find the sum of the interior angles of an octagon (8 sides).

a. 720°

b. $1,440^\circ$

c. 900°

d. $1,080^\circ$ ✓

$$(8 - 2) \times 180 = 1,080^\circ.$$

68 Find the measure of each interior angle of a regular hexagon.

- a. 140°
- b. 135°
- c. 108°
- d. 120° ✓

$$\text{Sum} = 720^\circ; \text{each} = 720 \div 6 = 120^\circ.$$

69 A polygon has an interior angle sum of $1,620^\circ$. Find the number of sides.

- a. 10
- b. 12
- c. 9
- d. 11 ✓

$$(n - 2) \times 180 = 1,620 \Rightarrow n - 2 = 9 \Rightarrow n = 11.$$

70 Find the sum of the interior angles of a decagon (10 sides).

- a. $1,800^\circ$
- b. $1,260^\circ$
- c. $1,080^\circ$
- d. $1,440^\circ$ ✓

$$(10 - 2) \times 180 = 1,440^\circ.$$

MCQ — Perimeter and area

71 A rectangle has length 15 and width 9, as shown. Find its perimeter and area.

a. $P = 135, A = 48$

b. $P = 48, A = 135$ ✓

c. $P = 24, A = 135$

d. $P = 48, A = 24$

$$P = 2(15 + 9) = 48; A = 15 \times 9 = 135.$$

72 A right triangle has legs 5 and 12, as shown. Find its area and hypotenuse.

a. $A = 60, \text{hyp} = 13$

b. $A = 30, \text{hyp} = 17$

c. $A = 30, \text{hyp} = 13$ ✓

d. $A = 30, \text{hyp} = 12$

$$A = \frac{1}{2}(5)(12) = 30; \text{hyp} = \sqrt{5^2 + 12^2} = 13.$$

73 A regular hexagon has side length 6, with area $A = \frac{3\sqrt{3}}{2}s^2$. Find its area in exact form.

a. $36\sqrt{3}$

b. $108\sqrt{3}$

c. $54\sqrt{3}$ ✓

d. $18\sqrt{3}$

$$A = \frac{3\sqrt{3}}{2}(36) = 54\sqrt{3}.$$

74 A regular pentagon has side length 8, with area $A \approx 1.72s^2$. Find its area to the nearest whole number.

a. 110 ✓

b. 124

c. 96

d. 14

$$A \approx 1.72(64) \approx 110.$$

MCQ — Coordinate geometry: identifying shapes

75 Is the triangle with vertices $A(0, 0)$, $B(6, 0)$, $C(6, 4)$ a right triangle?

- a. No
- b. Cannot be determined
- c. Yes ✓
- d. Only if it is also isosceles

AB is horizontal and BC is vertical, so they meet at a right angle at B : yes.

76 Determine whether the points $A(-3, 2)$, $B(1, 5)$, $C(4, 1)$ form a right triangle, using squared distances.

- a. Only if it is also equilateral
- b. No
- c. Cannot be determined
- d. Yes ✓

$AB^2 = 25$, $BC^2 = 25$, $AC^2 = 50$. Since $AB^2 + BC^2 = AC^2$, it's a right triangle at B : yes.

77 Determine whether the triangle with vertices $A(2, 3)$, $B(6, 3)$, $C(4, 7)$ is isosceles.

- a. No
- b. Only if it is also a right triangle
- c. Yes ✓
- d. Cannot be determined

$AC = \sqrt{4 + 16} = \sqrt{20}$ and $BC = \sqrt{4 + 16} = \sqrt{20}$: two equal sides, so yes.

78 Is the triangle shown a right triangle?

- a. Only if it is also isosceles
- b. Cannot be determined
- c. No
- d. Yes ✓

The two legs meet at a right angle, as shown.

79 Is the triangle shown isosceles, scalene, or equilateral?

- a. Isosceles ✓
- b. Cannot be determined
- c. Equilateral
- d. Scalene

Two of the three sides shown are equal in length: isosceles.

MCQ — The perpendicular bisector of a segment

80 Find the equation of the perpendicular bisector of the segment joining (1, 4) and (7, 10).

- a. $y = -x + 11$ ✓
- b. $y = x - 11$
- c. $y = x + 11$
- d. $y = -x + 7$

Midpoint = (4, 7); segment slope = 1; perpendicular slope = -1: $y - 7 = -(x - 4)$.

81 A line segment has endpoints (2, 0) and (2, 8). Find the equation of its perpendicular bisector.

a. $x = 4$

b. $y = 4$ ✓

c. $y = 2$

d. $x = 2$

Midpoint = (2, 4); the segment is vertical, so the bisector is horizontal: $y = 4$.

82 A line segment has endpoints (3, 6) and (9, 6). Find the equation of its perpendicular bisector.

a. $x = 3$

b. $y = 6$

c. $y = 9$

d. $x = 6$ ✓

Midpoint = (6, 6); the segment is horizontal, so the bisector is vertical: $x = 6$.

83 Find the equation of the perpendicular bisector of the segment joining (−2, 1) and (4, 9).

a. $y = -\frac{3}{4}x + \frac{23}{4}$ ✓

b. $y = -\frac{3}{4}x + 5$

c. $y = -\frac{4}{3}x + \frac{23}{4}$

d. $y = \frac{3}{4}x + \frac{23}{4}$

Midpoint = (1, 5); segment slope = $\frac{4}{3}$; perpendicular slope = $-\frac{3}{4}$.

MCQ — Word problems: geometry on a coordinate grid

- 84 A city park is designed on a coordinate grid (in meters). Two gates are located at (10, 20) and (50, 60). Find the distance between the gates, to the nearest meter.

a. 113

b. 40

c. 57 ✓

d. 80

$$\sqrt{40^2 + 40^2} = 40\sqrt{2} \approx 57 \text{ m.}$$

- 85 A circular irrigation sprinkler is centered at (8, 8) on a farm's coordinate grid (in meters) and has a radius of 15 meters. Write the equation of the circle it waters.

a. $(x + 8)^2 + (y + 8)^2 = 225$

b. $(x - 8)^2 + (y - 8)^2 = 15$

c. $(x - 8)^2 + (y - 8)^2 = 225$ ✓

d. $(x - 8)^2 + (y - 8)^2 = 30$

$$(x - 8)^2 + (y - 8)^2 = 15^2.$$

- 86 A surveyor marks two points on a coordinate map (in km): a base camp at (3, 4) and a summit at (15, 4). Find the midpoint, to be used as a resupply station.

a. (6, 4)

b. (9, 8)

c. (9, 4) ✓

d. (18, 8)

$$\left(\frac{3 + 15}{2}, \frac{4 + 4}{2} \right) = (9, 4).$$